

# Docker networking

Docker networking enables communication between containers, and between containers and the host or external systems. It provides various networking models that allow containers to connect securely and effectively, supporting diverse application needs.

## Key Concepts of Docker Networking:

1. **Container Networks:** Each container gets its own network namespace, including its own IP address, routing table, and network interfaces.
2. **Default Bridge Network:** By default, Docker creates a bridge network called `bridge`. When you start a container without specifying a network, it is attached to the default `bridge`.
3. **Network Drivers:** Docker provides different types of network drivers to suit various use cases.

# Types of networking:

## **Bridge Network** (Default):

- Used for single-host container communication.
- Containers connected to the bridge can communicate with each other.

## **Host Network:**

- The container shares the host's network stack.
- No IP isolation; the container uses the host's IP address.
- Suitable for performance-critical applications.

## **None Network:**

- No network is attached to the container.
- Used for highly secure containers without any network connectivity.

# Bridge Networking

```
[ec2-user@ip-172-31-84-217 ~]$ docker --version
```

```
Docker version 25.0.5, build 5dc9bcc
```

```
[ec2-user@ip-172-31-84-217 ~]$ docker images
```

| REPOSITORY | TAG    | IMAGE ID     | CREATED      | SIZE  |
|------------|--------|--------------|--------------|-------|
| tomcat     | latest | f62f518e5c5c | 9 days ago   | 467MB |
| nginx      | latest | 66f8bdd3810c | 3 weeks ago  | 192MB |
| httpd      | latest | 494b2b45fd74 | 5 months ago | 147MB |

```
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --name web -p 8080:80 httpd
```

```
a08da5222987acba86b1dc26c67b74c652ecc9b35d38a8e91084eb39c6a1a22a
```

```
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --name web1 --net bridge -p 8081:80 nginx
```

```
edbaed90665a9a1c00da09c327b6c1b9015a9f6a07c01b9861c35a0c8416b5e6
```

```
[ec2-user@ip-172-31-84-217 ~]$ docker ps
```

| CONTAINER ID | IMAGE | COMMAND                 | CREATED        | STATUS        | PORTS                                 |
|--------------|-------|-------------------------|----------------|---------------|---------------------------------------|
| edbaed90665a | nginx | "/docker-entrypoint..." | 20 seconds ago | Up 20 seconds | 0.0.0.0:8081->80/tcp, :::8081->80/tcp |
| cp web1      |       |                         |                |               |                                       |
| a08da5222987 | httpd | "httpd-foreground"      | 39 seconds ago | Up 38 seconds | 0.0.0.0:8080->80/tcp, :::8080->80/tcp |
| cp web       |       |                         |                |               |                                       |

```
[ec2-user@ip-172-31-84-217 ~]$
```

# Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

## Inbound rules [Info](#)

| Security group rule ID | Type <a href="#">Info</a> | Protocol <a href="#">Info</a> | Port range <a href="#">Info</a> | Source <a href="#">Info</a> | Description - optional <a href="#">Info</a> |                                       |
|------------------------|---------------------------|-------------------------------|---------------------------------|-----------------------------|---------------------------------------------|---------------------------------------|
| sgr-034ae147851c7fe0c  | SSH ▼                     | TCP                           | 22                              | Cu... ▼                     | <input type="text" value="0.0.0.0/0"/>      | <input type="button" value="Delete"/> |
| -                      | Custom TCP ▼              | TCP                           | 8080                            | An... ▼                     | <input type="text" value="0.0.0.0/0"/>      | <input type="button" value="Delete"/> |
| -                      | Custom TCP ▼              | TCP                           | 8081                            | An... ▼                     | <input type="text" value="0.0.0.0/0"/>      | <input type="button" value="Delete"/> |

Add rule

⚠ Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.





**It works!**

# Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to [nginx.org](https://nginx.org).  
Commercial support is available at [nginx.com](https://nginx.com).

*Thank you for using nginx.*

# Host networking:

```
[ec2-user@ip-172-31-84-217 ~]$ docker ps -a
```

| CONTAINER ID | IMAGE | COMMAND | CREATED | STATUS | PORTS | NAMES |
|--------------|-------|---------|---------|--------|-------|-------|
|--------------|-------|---------|---------|--------|-------|-------|

```
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --name web2 --net host httpd
```

```
def8689b6c93be4e22ef53e398d019c009306b13e2970d29e8433687f3812a88
```

```
[ec2-user@ip-172-31-84-217 ~]$ docker ps
```

| CONTAINER ID | IMAGE | COMMAND            | CREATED        | STATUS        | PORTS | NAMES |
|--------------|-------|--------------------|----------------|---------------|-------|-------|
| def8689b6c93 | httpd | "httpd-foreground" | 18 seconds ago | Up 17 seconds |       | web2  |

```
[ec2-user@ip-172-31-84-217 ~]$
```

```
"SecondaryIPv6Addresses": null,  
"EndpointID": "",  
"Gateway": "",  
"GlobalIPv6Address": "",  
"GlobalIPv6PrefixLen": 0,  
"IPAddress": "",  
"IPPrefixLen": 0,  
"IPv6Gateway": "",  
"MacAddress": "",  
"Networks": {  
  "host": {  
    "IPAMConfig": null,  
    "Links": null,  
    "Aliases": null,  
    "MacAddress": "",  
    "NetworkID": "ec480e22fa3c8ef66624d45fc9dc43b0b217e5fbe818c8ff80c6f72b63598f07",  
    "EndpointID": "dde0c6a3299068a7ebd79d9ab86554829a88c97a90867bf4e7ec6e451b3306d4",  
    "Gateway": "",  
    "IPAddress": "",  
    "IPPrefixLen": 0,  
    "IPv6Gateway": "",  
    "GlobalIPv6Address": "",  
    "GlobalIPv6PrefixLen": 0,  
    "DriverOpts": null,  
    "DNSNames": null  
  }  
}  
}  
}  
}  
]  
[ec2-user@ip-172-31-84-217 ~]$
```



# Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

## Inbound rules

| Security group rule ID | Type        | Protocol | Port range | Source | Description - optional |        |
|------------------------|-------------|----------|------------|--------|------------------------|--------|
| sgr-034ae147851c7fe0c  | SSH         | TCP      | 22         | Cu...  |                        | Delete |
| -                      | All traffic | All      | All        | An...  |                        | Delete |

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

Preview changes

Save rules



**It works!**

# None network

```
[ec2-user@ip-172-31-84-217 ~]$ docker ps
```

| CONTAINER ID | IMAGE | COMMAND            | CREATED       | STATUS       | PORTS | NAMES |
|--------------|-------|--------------------|---------------|--------------|-------|-------|
| def8689b6c93 | httpd | "httpd-foreground" | 4 minutes ago | Up 4 minutes |       | web2  |

```
[ec2-user@ip-172-31-84-217 ~]$ docker run -itd --name web --net none nginx
```

```
af3e0de4d0b3c2ef6ca355b8cbdd3183dfcf33752f08160edb428714a8814731
```

```
[ec2-user@ip-172-31-84-217 ~]$ docker ps
```

| CONTAINER ID | IMAGE | COMMAND                 | CREATED        | STATUS        | PORTS | NAMES |
|--------------|-------|-------------------------|----------------|---------------|-------|-------|
| af3e0de4d0b3 | nginx | "/docker-entrypoint..." | 16 seconds ago | Up 15 seconds |       | web   |
| def8689b6c93 | httpd | "httpd-foreground"      | 5 minutes ago  | Up 5 minutes  |       | web2  |

```
[ec2-user@ip-172-31-84-217 ~]$
```

```
"SecondaryIPv6Addresses": null,
"EndpointID": "",
"Gateway": "",
"GlobalIPv6Address": "",
"GlobalIPv6PrefixLen": 0,
"IPAddress": "",
"IPPrefixLen": 0,
"IPv6Gateway": "",
"MacAddress": "",
"Networks": {
  "none": {
    "IPAMConfig": null,
    "Links": null,
    "Aliases": null,
    "MacAddress": "",
    "NetworkID": "5960f0276ec834bdbfe097cacdf08eb943cd5b4e92fb0de26",
    "EndpointID": "d0ea41e43b539beb60ebc4ae8b282141c935c92ebb37871a741d14fb2c604c4d",
    "Gateway": "",
    "IPAddress": "",
    "IPPrefixLen": 0,
    "IPv6Gateway": "",
    "GlobalIPv6Address": "",
    "GlobalIPv6PrefixLen": 0,
    "DriverOpts": null,
    "DNSNames": null
  }
}
}
```

]

[ec2-user@ip-172-31-84-217 ~]\$



34.201.59.201



## This site can't be reached

**34.201.59.201** refused to connect.

Try:

- Checking the connection
- [Checking the proxy and the firewall](#)

ERR\_CONNECTION\_REFUSED

Reload

Details